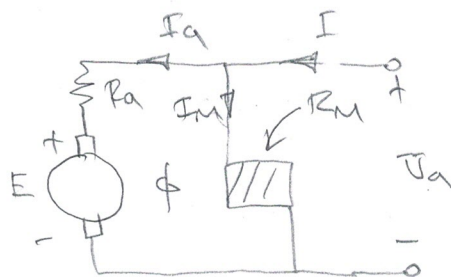


19.7)



$$\begin{aligned} R_M &= 400 \, \Omega \\ R_a &= 0.2 \, \Omega \\ U_a &= 440 \, \text{V} \\ n_0 &= 1200 \, \text{rpm} \\ P &= 100 \, \text{HP} \\ \eta &= 90\% \, \text{TOTAL} \end{aligned}$$

a) Beräkna strömmen I , storsta effekter

$$\begin{aligned} P_M &= \eta P_{IN} \Rightarrow P_{IN} = \frac{1}{\eta} P_M = \frac{1}{0.9} P_0 = \\ &= \frac{1}{0.9} 100 \cdot 745.7 = \underline{82855.6 \, \text{W}} \end{aligned}$$

MEN ELEKTISK EFFEKT IN MÅSTE JÄMFÖRAS
AV MEKANISK EFFEKT $U_a I$, DVS

$$\begin{aligned} U_a I &= P_{IN} \Rightarrow I = \frac{P_{IN}}{U_a} = \\ &= \frac{82855.6}{440} = \underline{188.3 \, \text{A}} \end{aligned}$$

b) I_M och I_a ?

$$\begin{aligned} I_M &= \frac{440}{400} = 1.1 \, \text{A} \Rightarrow I_a = I - I_M = \\ &= 188.3 - 1.1 = \\ &= \underline{187.2 \, \text{A}} \end{aligned}$$

c) E ?

$$\text{KSP LÖSER } -U_a + R_a I_a + E = 0$$

$$E = U_a - R_a I_a =$$

$$= 440 - 0.2 \cdot 187.2 = \underline{402.6 \, \text{VOLT}}$$

d) VRIDMOMENT?

$$P_M = M \omega \Rightarrow M = \frac{P_M}{\omega} = \frac{100 \cdot 745.7}{1200 \cdot \frac{2\pi}{60}} =$$

$$\underline{\text{SVAR: a) } I = 188.3 \, \text{A} \quad \quad \quad = 593.41 \, \text{NM}}$$

$$\text{b) } I_a = 1.1 \, \text{A}, I_M = 187.2 \, \text{A}$$

$$\text{c) } E = 402.6 \, \text{V}$$

$$\text{d) } M = 593.4 \, \text{NM}$$